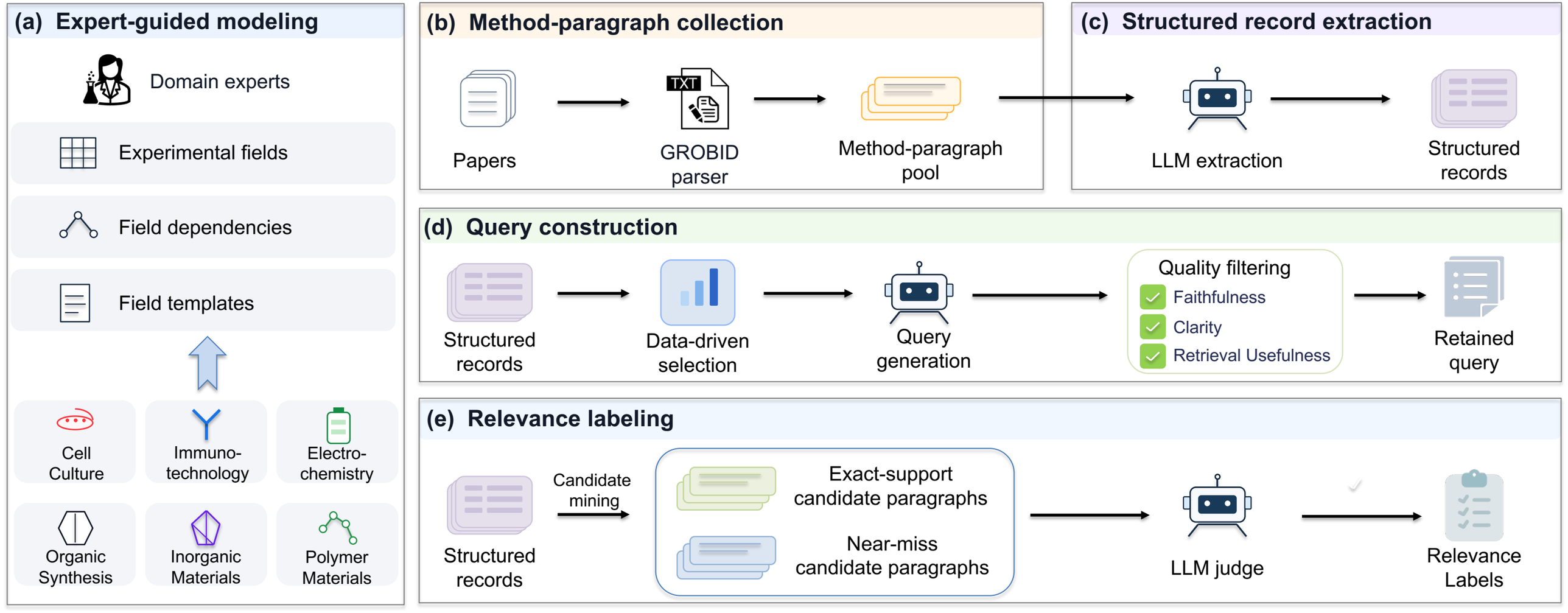

(a) Expert-guided modeling
Domain experts
Experimental fields
Field dependencies
Field templates
Cell Culture
Immuno-technology
Electro-chemistry
Organic Synthesis
Inorganic Materials
Polymer Materials
(b) Method-paragraph collection
Papers
TXT
GROBID parser
Method-paragraph pool
(c) Structured record extraction
LLM extraction
Structured records
(d) Query construction
Structured records
Data-driven selection
Query generation
Quality filtering
Faithfulness
Clarity
Retrieval Usefulness
Retained query
(e) Relevance labeling
Structured records
Candidate mining
Exact-support candidate paragraphs
Near-miss candidate paragraphs
LLM judge
Relevance Labels